

Annual Trading Statistics of the Colombo Stock Exchange: An Explanatory Statistical Analysis

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INTRODUCTION

The Colombo Stock Exchange (CSE) plays a vital role in Sri Lanka's financial system by facilitating capital formation and providing a regulated platform for equity trading. Trading activity in stock markets is commonly assessed using indicators such as turnover, trading volume, and price volatility, which together provide insights into market liquidity, investor participation, and overall market conditions. Understanding the behavior of these indicators over time is particularly important in emerging markets, where market dynamics may be influenced by structural changes, varying economic conditions, and shifts in investor sentiment.

This study focuses on the analysis of **annual trading statistics of selected listed companies across different GICS (Global Industry Classification Standard) industry groups** in the Colombo Stock Exchange. The dataset, obtained from the official Colombo Stock Exchange website, covers the period from 2006 to 2022 and includes key trading-related variables such as turnover, share volume, trade volume, price volatility, and market phase.

<https://www.cse.lk/equity/annual-trading-statistics>

Problem Definition

The statistical problem addressed in this study is to **examine long-term changes in annual trading activity and market performance from 2006 to 2022 for selected listed companies in the Colombo Stock Exchange**. Specifically, the analysis investigates how turnover is associated with price volatility (measured by the high-low price range), share volume, trade volume, and market phase. Given the presence of potential outliers and non-uniform trading patterns in financial data, the study emphasizes explanatory analysis and statistical inference rather than prediction.

Objectives

- ♣ Explore the key characteristics and behavior of trading turnover and related market variables using exploratory data analysis.
- ♣ Assess whether different market phases have a statistically significant effect on turnover growth.
- ♣ Examine the relationship between price volatility and turnover, share volume, and trade volume for the selected companies.
- ♣ Apply robust statistical modeling techniques to explain turnover behavior and support conclusions using appropriate statistical inference.

DATA DESCRIPTION & PREPROCESSING

Before analysis, the availability of data for all required years within this period was verified to ensure consistency of the time span.

To facilitate analysis and ensure consistency, several preprocessing steps were applied. Column names were standardized by replacing spaces with underscores, removing full stops, and trimming leading or trailing whitespace. These steps were necessary to enable smooth data manipulation and analysis using Python-based statistical tools.

The initial dataset contained the following variables:

Variable	Description
Company_Name	Full legal name of the listed company
Symbol	Unique short code identifying the company on the stock exchange
Open_(Rs)	Opening price of the stock in a given year
High_(Rs)	The highest price at which the stock traded during the year
Low_(Rs)	The lowest price at which the stock traded during the year
Close_(Rs)	Closing price of the stock in a given year
Trade_Volume	Total number of trades executed during the year
Share_Volume	Total number of shares traded during the year
Turnover(Rs)	Total monetary value of shares traded during the year

Company symbols were standardized by converting them to uppercase string format and removing any embedded spaces, ensuring consistency across different data sources and years.

Under annual trading statistics, it is expected that each company would have a single record per year. However, the dataset contained multiple records for certain companies within the same year, with differing values for trading variables. This can reasonably be attributed to operational practices of the CSE, such as:

- ♣ **Market segmentation:** Companies may trade on different boards (e.g., Main Board or Diri Savi Board), resulting in separate records if a transfer occurs during the year.
- ♣ **Suspension and re-listing:** Temporary suspension and subsequent re-listing under revised conditions may lead to the reporting of trading activity as multiple entries.

To address this issue, such records were aggregated appropriately at the company-year level, ensuring that each company had a single consolidated annual record suitable for longitudinal analysis.

```
# Aggregation
agg_dict = {
    'Company_Name': 'first',
    'Open_(Rs)': 'mean',
    'High_(Rs)': 'max',
    'Low_(Rs)': 'min',
    'Close_(Rs)': 'mean',
    'Trade_Volume': 'sum',
    'Share_Volume': 'sum',
    'Turnover_(Rs)': 'sum'
}
```

The number of listed companies varied over time, indicating that not all companies were continuously listed throughout the period from 2006 to 2022. To maintain a consistent longitudinal framework, only companies that were present throughout the selected time period were retained. This filtering resulted in a total of **268** companies.

Given the large number of companies, it was not practical to analyze all firms individually. To ensure a meaningful comparison, companies were grouped by GICS (Global Industry Classification Standard) industry groups. Since the original dataset did not contain GICS classifications, this information was manually obtained from the CSE website and merged into the main dataset.

The original data were provided in separate worksheets for each year. To simplify analysis and enable time-based modeling, all annual data were consolidated into a single dataset, including a new variable, **Year**, indicating the corresponding observation year. The resulting dataset was saved as an Excel file named *final_merged_dataset*.

The consolidated dataset comprised **20** distinct GICS industry groups. From each group, the company with the highest turnover over the study period was identified. Among these, companies exhibiting distinct and informative trading patterns were selected for detailed analysis. The final selected companies were:

- Commercial Bank of Ceylon PLC
- Softlogic Life Insurance PLC
- LOLC Holdings PLC
- Expolanka Holdings PLC

These companies represent different industry groups and display unique turnover and trading dynamics, making them suitable for in-depth statistical analysis.

In addition, price volatility was measured using the annual high–low price range, and turnover was log-transformed in subsequent analysis to reduce skewness and stabilize variance.

➤ During preliminary data inspection, it was observed that the annual trading statistics for the years 2010–2011 and 2020–2021 are identical across all variables for the selected companies. This may be attributed to a combination of data aggregation practices and exceptional market conditions. In particular, the 2020–2021 period coincides with COVID-19–related market disruptions and regulatory constraints, which likely reduced variability in annual trading activity. Similarly, the similarity observed in 2010–2011 may reflect post-conflict market stabilization following the end of the civil war, together with the aggregated nature of the Annual Trading Statistics dataset.

As these values are directly sourced from the official Annual Trading Statistics, they were retained in the analysis and interpreted with caution.

COMMERCIAL BANK OF CEYLON PLC

Commercial Bank of Ceylon PLC is listed on the Colombo Stock Exchange under two different symbols, COMB.N0000 and COMB.X0000, which represent voting and non-voting ordinary shares, respectively. While both share classes provide similar financial benefits, such as dividends, only the voting shares carry shareholder voting rights, resulting in differences in liquidity and trading activity. For the purpose of this analysis, **COMB.N0000** was selected as it recorded the highest turnover within the Banking sector over the study period.

	Open_(Rs)	High_(Rs)	Low_(Rs)	Close_(Rs)	Trade_Volume	Share_Volume	Turnover(Rs)
count	17.000000	17.000000	17.000000	17.000000	17.000000	1.700000e+01	1.700000e+01
mean	109.883824	158.820588	58.135294	97.423529	81646.588235	3.848227e+08	3.120946e+10
std	17.717633	64.257585	24.068780	29.819237	57212.570227	2.531017e+08	1.876196e+10
min	79.100000	85.000000	48.700000	50.200000	11811.000000	9.614552e+07	6.470299e+09
25%	97.500000	115.900000	50.000000	80.900000	38690.000000	2.037215e+08	1.855291e+10
50%	117.125000	149.750000	50.000000	90.450000	91073.000000	3.969488e+08	2.969530e+10
75%	117.125000	192.000000	50.000000	125.950000	99181.000000	4.486606e+08	4.018135e+10
max	144.000000	284.000000	128.500000	145.000000	257742.000000	1.155050e+09	8.327741e+10

Before conducting further exploratory analysis, descriptive statistics were computed to summarize the central tendency, dispersion, and range of key trading variables for COMB.N0000.

The price-related variables (Open, High, Low, and Close) indicate notable variability across years, reflecting fluctuations in market conditions. The average share volume traded is approximately 3.85×10^8 shares, with a relatively high standard deviation, indicating substantial variation in investor participation across years. Turnover shows a high mean value of Rs. 3.12×10^{10} , accompanied by considerable dispersion. Overall, the descriptive statistics confirm that COMB.N0000 exhibits sufficient variability in prices and trading activity, justifying its suitability for further exploratory analysis focusing on turnover, volume, and volatility.

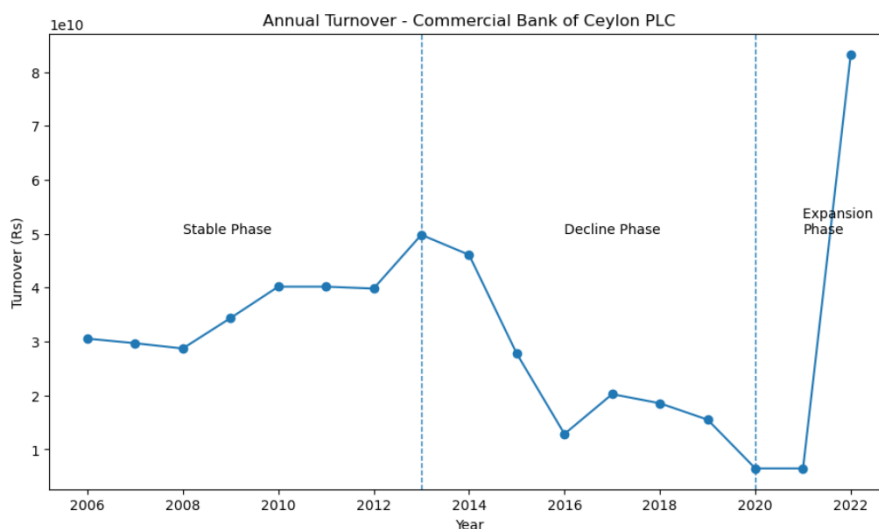


Figure 01

Figure 01 illustrates the year-to-year movement in turnover of Commercial Bank of Ceylon PLC (COMB.N0000) from 2006 to 2022. The dashed vertical lines mark structural breaks in turnover behavior, separating the time period into distinct market phases, based on observed changes in the level and trend of turnover.

Stable Phase (2006 - 2013):

Turnover remained relatively stable with a gradual upward trend. Although minor fluctuations are observed, turnover generally increases from approximately Rs. 3.0×10^{10} to Rs. 5.0×10^{10} . This indicates consistent investor participation and stable liquidity in the stock over this period.

Decline Phase (2014 - 2019):

A sharp decline in turnover is observed from 2014 onwards, marking a clear structural shift in trading activity. Turnover falls substantially, reaching its lowest levels around 2016 and again in 2019. This period is characterized by reduced liquidity and weaker investor participation.

Expansion Phase (2020 - 2022):

From 2020 onwards, turnover initially remains low but then increases sharply in 2022, reaching its highest value in the entire study period. This sudden surge indicates a strong expansion phase characterized by renewed investor interest, increased trading activity, and heightened market participation.

Overall, the plot reveals that turnover does not follow a uniform trend over time but instead exhibits distinct phases of stability, decline, and expansion. This highlights the importance of examining turnover dynamics across different market conditions rather than relying solely on aggregate statistics.

While the level of turnover provides valuable insight into market liquidity, it does not capture the rate at which trading activity changes over time. Therefore, to better understand the dynamics of market participation across different phases, year-to-year turnover growth rates are analyzed. Examining turnover growth allows for the identification of periods of acceleration, contraction, and recovery, and enables a clearer comparison of market behavior across the identified phases.

Figure 02 presents the year-to-year growth rate of turnover. The growth rates fluctuate around zero during the early years, indicating relatively stable trading activity. From 2014 onwards, the growth rates are largely negative, reflecting a sustained decline in turnover during this period. A clear turning point is observed around 2020, after which turnover growth becomes positive. The exceptionally large growth rate observed in 2022 represents a structural shock, corresponding to a sharp surge in trading activity and marking a strong expansion phase.

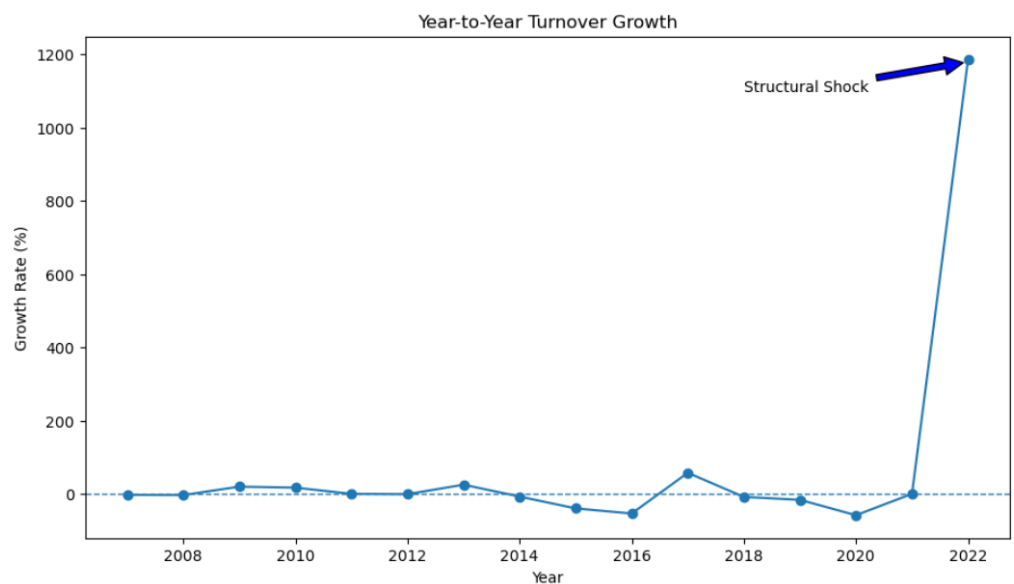


Figure 02

To further quantify the differences in turnover dynamics across the identified phases (Stable, Decline, Expansion), summary statistics of the year-to-year turnover growth rates were computed for each phase.

Phase	mean	std	min	max	count
Phase 1: Stable	4.941637	10.454260	-3.302032	19.662172	6
Phase 2: Decline	-6.167690	37.680852	-53.678796	57.456233	7
Phase 3: Expansion	376.260895	702.787630	-58.289361	1187.072044	3

observed growth is not uniform but dominated by a structural shock rather than steady expansion.

While the phase-wise summary statistics reveal notable differences in average growth rates and variability across phases, a formal statistical test is required to assess whether these differences are statistically significant. Given the non-normal distribution of turnover growth rates and the presence of extreme values, a Kruskal–Wallis test was employed to compare the distributions of turnover growth across the identified market phases.

Kruskal-Wallis Test

H_0 : The distribution of turnover growth rates is the same across all market phases.

H_1 : At least one market phase has a significantly different turnover growth distribution.

The Kruskal–Wallis test results indicate that there is **no statistically significant difference in the distribution of turnover growth rates across the three market phases** ($H = 1.287$, $p = 0.5255$) at 5% significance level. Although the descriptive and graphical analyses suggest distinct phase-wise patterns in turnover behavior, the lack of statistical significance indicates that the observed differences in growth rates are largely driven by variability within phases, particularly the presence of extreme observations during the expansion phase.

- The limited number of observations within each phase and the influence of structural shocks may have reduced the power of the test to detect statistically significant differences.

While turnover and growth capture trading intensity and changes in market participation, they do not reflect the degree of price uncertainty faced by investors. Therefore, volatility is analyzed to examine fluctuations in price movements over time.

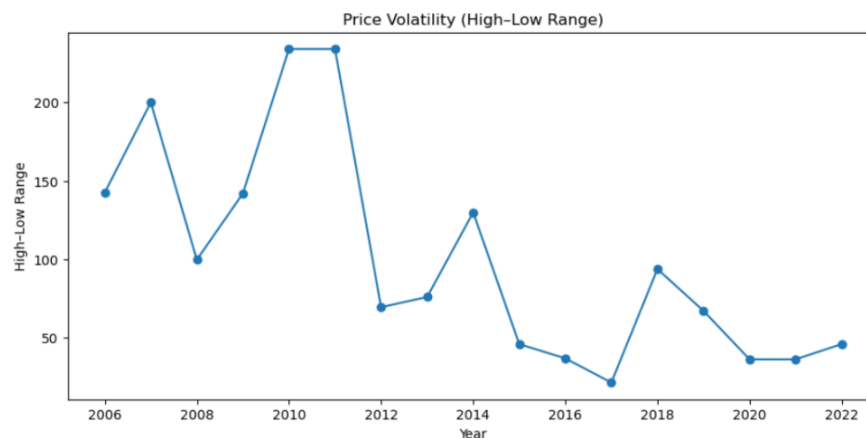
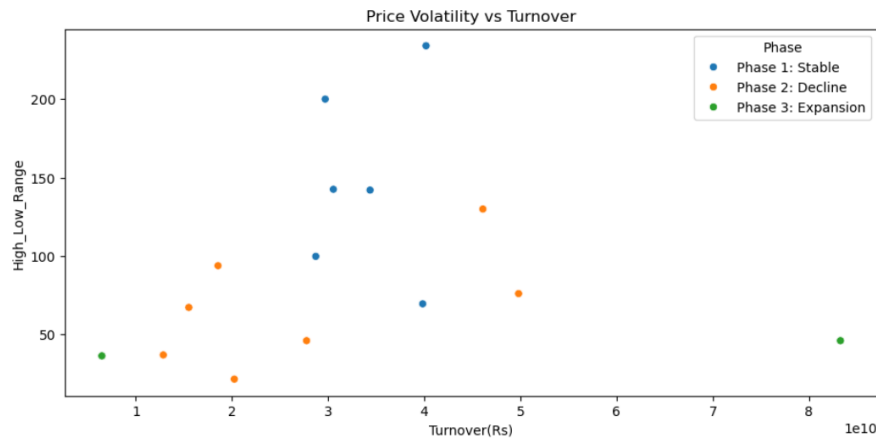


Figure 03

The annual volatility values exhibit substantial variation across the study period, indicating changes in market uncertainty over time. Higher volatility values observed in earlier years suggest larger price fluctuations and increased uncertainty, while the noticeably lower volatility in later years indicates a more stable price environment.



When examined alongside the previously identified turnover phases, higher volatility is generally observed during periods of elevated trading activity and structural change, while lower volatility corresponds to phases of reduced market participation. This reinforces the view that volatility dynamics differ across market regimes.

Following the analysis of price volatility, correlation analysis was conducted to examine the relationship between volatility and key trading activity measures, namely turnover, share volume, and trade volume.

Turnover(Rs):

Pearson correlation = 0.251
Spearman correlation = 0.565

Share_Volume:

Pearson correlation = 0.190
Spearman correlation = 0.553

Trade_Volume:

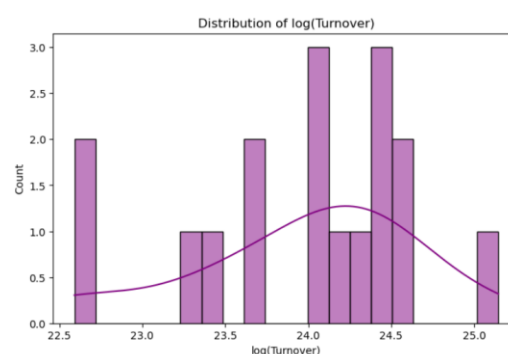
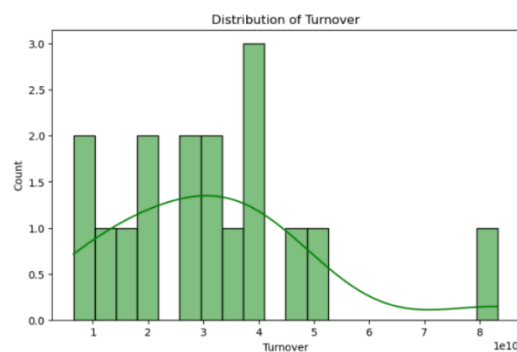
Pearson correlation = 0.230
Spearman correlation = 0.529

The correlation results indicate a positive association between price volatility and all three trading activity measures. Pearson correlations are relatively weak, ranging from 0.19 to 0.25, suggesting limited linear relationships. In contrast, Spearman correlations are moderately strong, with values above 0.52 for all variables, indicating a stronger monotonic relationship between trading activity and volatility.

The stronger Spearman correlations relative to Pearson correlations suggest that the relationship between volatility and trading activity is not strictly linear. Instead, volatility tends to increase with higher

levels of turnover and trading volume in a non-linear manner, particularly during periods of heightened market activity. Among the variables considered, turnover exhibits the strongest association with volatility, followed closely by share volume and trade volume. This implies that periods of increased liquidity and trading intensity are generally accompanied by higher price fluctuations.

While correlation analysis provides insight into pairwise associations between volatility and trading activity, it does not account for the combined effect of multiple variables or the presence of extreme observations. Therefore, a multivariate regression framework was employed to examine the relationship between turnover, volatility, and trading activity while controlling for market phases.



Exploratory analysis revealed that annual turnover exhibited significant right skewness. To address this and improve model stability, turnover was log-transformed. The log-transformed series displayed a more symmetric distribution, making it more suitable for regression analysis and interpretation.

Robust Linear Model

Given the presence of extreme observations and structural shocks in turnover, a robust linear regression model using Huber's M-estimator was fitted. This approach reduces the influence of outliers while retaining efficiency for approximately normally distributed errors.

$$\log(\text{Turnover}) = 23.0312 + 0.0011(\text{High-Low Range}) + 4.261 \times 10^{-9}(\text{Share Volume}) - 8.088 \times 10^{-6}(\text{Trade Volume}) - 0.0571(\text{Phase 2: Decline}) - 0.7672(\text{Phase 3: Expansion}) + \epsilon$$

Price Volatility (High-Low Range): The coefficient on price volatility is positive and statistically significant ($p = 0.001$), indicating that higher price fluctuations are associated with increased turnover.

Share Volume: Share volume exhibits a positive and statistically significant relationship ($p < 0.001$), implying that higher quantities of shares traded contribute directly to increased market value traded.

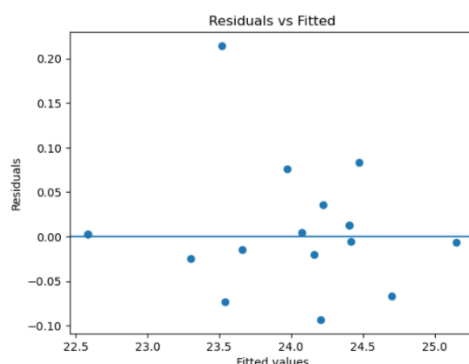
Trade Volume: Trade volume shows a statistically significant negative coefficient ($p = 0.001$), suggesting that, conditional on total shares traded, a higher number of trades may be associated with smaller average trade sizes.

Phase 2: Decline: The coefficient for the decline phase is negative but not statistically significant, indicating that turnover during this period does not differ significantly from the stable phase once volatility and trading activity are accounted for.

Phase 3: Expansion: The expansion phase exhibits a large and statistically significant negative coefficient ($p < 0.001$). This reflects the influence of extreme turnover values and structural shocks during this period, even after log transformation.

It should be noted that the model captures associations rather than causal relationships, and results should be interpreted within the context of exploratory financial analysis.

Overall, the robust regression analysis demonstrates that turnover is jointly influenced by price volatility, trading intensity, and market regime, reinforcing the interconnected nature of liquidity, risk, and market structure.



The residuals versus fitted values plot shows no discernible pattern, with residuals scattered randomly around zero. This suggests the absence of strong non-linear relationships or heteroscedasticity that would undermine the interpretation of the estimated coefficients.

SOFTLOGIC LIFE INSURANCE PLC

Descriptive Analysis

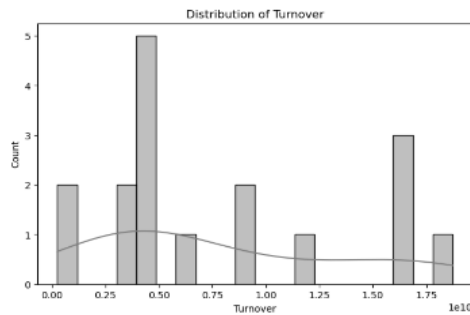
	Open_(Rs.)	High_(Rs.)	Low_(Rs.)	Close_Price_(Rs)	Trade_Volume	Share_Volume	Turnover(Rs)	Year
count	17.000000	17.000000	17.000000	17.000000	17.000000	1.700000e+01	1.700000e+01	17.000000
mean	38.189706	151.526471	19.588235	56.723529	46475.352941	1.348653e+08	7.807469e+09	2014.000000
std	19.321996	129.957827	5.127241	27.967249	62544.434948	6.995431e+07	5.909848e+09	5.049752
min	16.000000	24.000000	7.000000	20.700000	2950.000000	1.316081e+07	2.539520e+08	2006.000000
25%	23.675000	38.600000	20.000000	34.800000	9955.000000	1.106401e+08	4.030213e+09	2010.000000
50%	23.675000	107.000000	20.000000	60.150000	12723.000000	1.175449e+08	4.164375e+09	2014.000000
75%	57.500000	249.000000	20.000000	86.600000	27975.000000	1.554747e+08	1.180432e+10	2018.000000
max	70.000000	405.000000	27.000000	100.750000	159642.000000	3.266771e+08	1.872542e+10	2022.000000

The data covers 17 years (2006–2022), enabling long-term analysis. The average closing price (Rs.56.72) is higher than the opening price (Rs.38.19), indicating overall price growth. High prices show wide fluctuations, ranging from Rs.24 to Rs.405.

Lower prices remain relatively stable around Rs.20, suggesting a support level. Trading volume and turnover vary considerably, indicating changing investor interest while maintaining overall good liquidity.

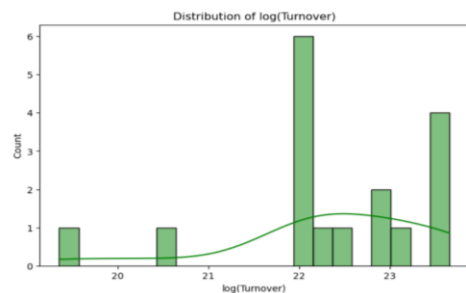
Exploratory Analysis

Distribution of Turnover



The company's turnover is unevenly distributed, with most observations concentrated at lower levels and a few extremely high values, resulting in a right-skewed distribution. These occasional spikes can disproportionately influence the average turnover. Overall, turnover varies considerably over time rather than following a consistent growth trend.

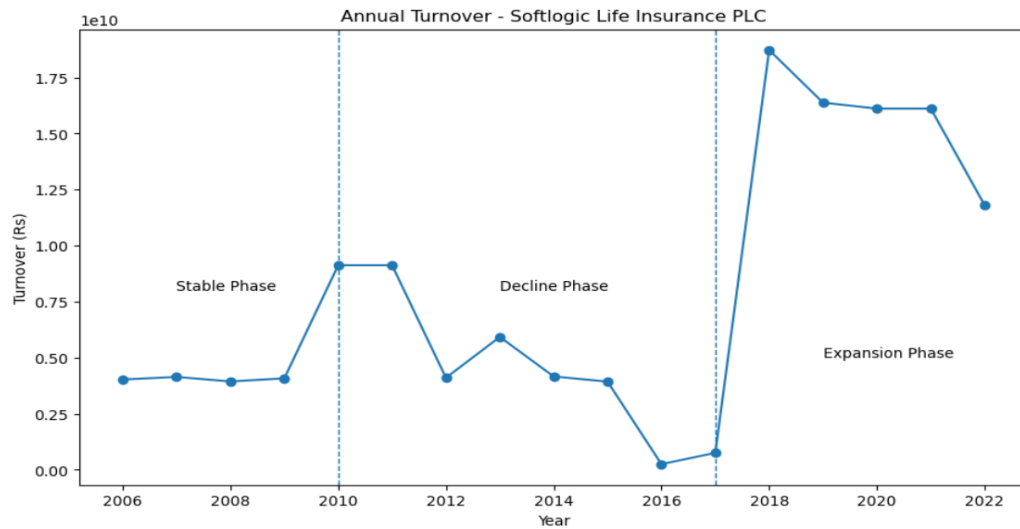
Distribution of log(Turnover)



The distribution of log-transformed turnover is slightly left-skewed, suggesting that most turnover values are relatively high, while a smaller number of lower values stretch the distribution toward the lower end. This indicates that although high turnover is common, occasional periods of unusually low turnover occur, contributing to the left tail of the distribution.

The original turnover distribution was slightly right-skewed due to a few extremely high values. A logarithmic transformation was applied to reduce skewness. Although the log-transformed turnover shows slight left skewness, it provides a more balanced representation of the data and reduces the influence of extreme observations.

Annual Turnover

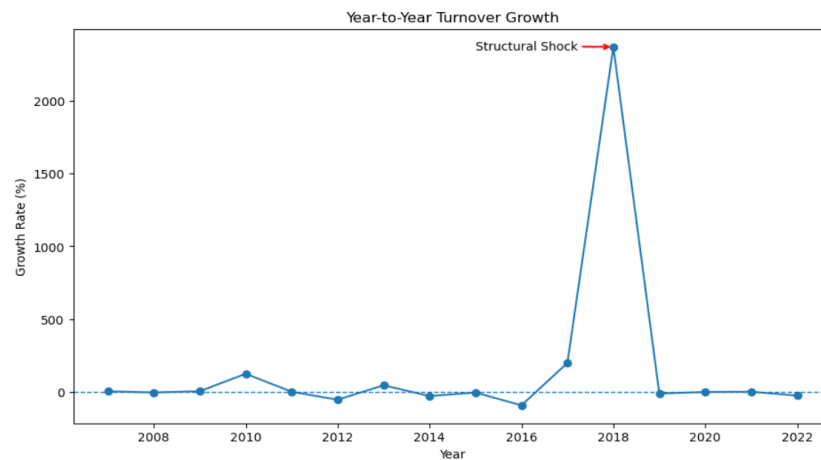


Based on the annual turnover plot, the period under study can be divided into three distinct phases. The first phase (2006–2010) shows relatively stable turnover levels, followed by a second phase (2011–2017) characterized by a gradual decline. The final phase, beginning in 2017, exhibits a sharp increase in turnover, indicating a transition from stability to decline and subsequently to strong growth.

Phase	mean	std	min	max	count
Phase 1: Stable	25.016533	55.205797	-4.841733	123.596649	5
Phase 2: Decline	9.797268	103.324267	-93.545956	198.225335	6
Phase 3: Expansion	466.319700	1065.637571	-26.736439	2372.494348	5

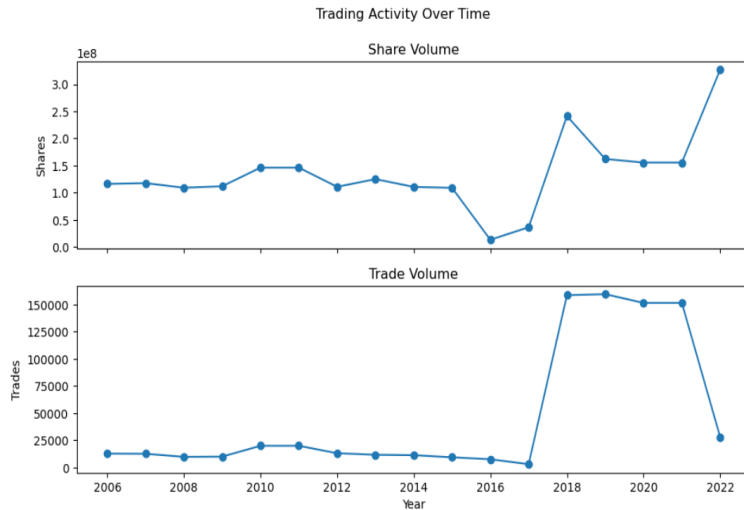
Phase 1 (Stable) shows low values with moderate fluctuations. **Phase 2 (Decline)** has lower averages but high volatility and big downturns. **Phase 3 (Expansion)** shows very high values with large swings, reflecting strong but volatile growth.

Year-to-Year Turnover Growth



Year-to-year turnover growth is generally moderate, except for a significant spike in 2018. According to the company's 2018 annual report, this increase resulted from a one-off accounting change in 2017. As the effect was not sustained, turnover growth normalized in the years that followed.

Trading Activity over Time



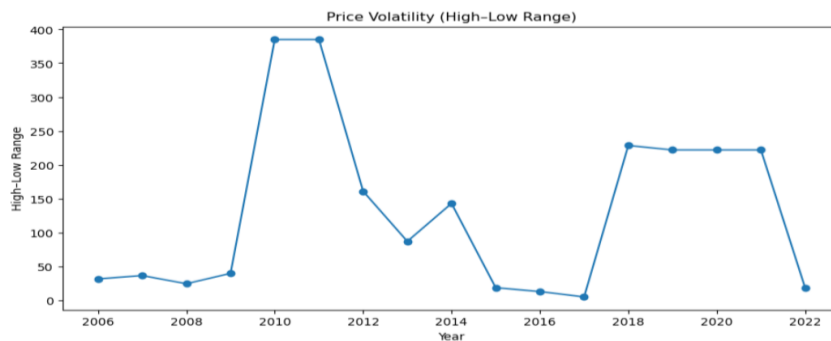
Share Volume:

Share volume remains relatively stable in the early years and increases noticeably from 2017 onwards.

Trade Volume:

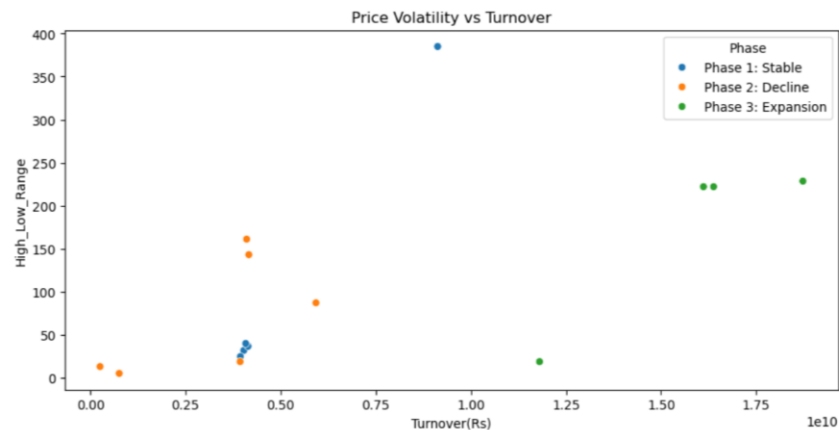
Trade volume stays low and stable for most of the period, shows a sharp rise around 2018, remains elevated for several subsequent years, and then declines again.

Price Volatility over Time



Price volatility remains low and stable in the early years, reflecting relatively calm market conditions. It rises sharply during 2009–2011, likely in response to heightened market uncertainty, before declining after 2014 as stability returns. Volatility increases again between 2018 and 2021, possibly driven by renewed market shocks or increased trading activity, and falls in the most recent year, indicating a return to more stable conditions.

Price Volatility vs Turnover



Phase 1 (Stable) has low turnover and low volatility. **Phase 2 (Growth)** shows moderate turnover with more fluctuation. **Phase 3 (Expansion)** has the highest turnover and high volatility. Overall, both turnover and volatility increase as the company moves from stability to expansion.

Correlation

Turnover(Rs):

Pearson correlation = 0.597

Spearman correlation = 0.772

Share_Volume:

Pearson correlation = 0.268

Spearman correlation = 0.629

Trade_Volume:

Pearson correlation = 0.448

Spearman correlation = 0.750

Turnover shows the strongest positive relationship with price volatility (Pearson: 0.597, Spearman: 0.772), indicating that higher trading intensity is associated with higher volatility. Trade volume also exhibits a moderate to strong positive association with volatility (Pearson: 0.448, Spearman: 0.750). In contrast, share volume has a weaker linear relationship with volatility (Pearson: 0.268), suggesting it is a less informative measure of volatility.

Hypothesis testing

H_0 : The distribution of turnover growth rates is the same across all market phases.

H_1 : At least one market phase has a significantly different turnover growth distribution.

Kruskal-Wallis H statistic: 1.960

P-value: 0.3753

P-value (0.3753) > 0.05, Fail to reject the null hypothesis.

This indicates no statistically significant difference in turnover growth distributions across the market phases. Turnover growth rates are similar across all market phases.

Model Building

OLS Regression Results						
=====						
Dep. Variable:	Turnover(Rs)	R-squared:	0.481			
Model:	OLS	Adj. R-squared:	0.407			
Method:	Least Squares	F-statistic:	6.486			
Date:	Sat, 31 Jan 2026	Prob (F-statistic):	0.0102			
Time:	10:26:01	Log-Likelihood:	-19.972			
No. Observations:	17	AIC:	45.94			
Df Residuals:	14	BIC:	48.44			
Df Model:	2					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	22.2842	0.386	57.722	0.000	21.456	23.112
Phase2	-0.6251	0.505	-1.237	0.237	-1.709	0.459
Phase3	1.1897	0.546	2.179	0.047	0.019	2.361
=====						
Omnibus:	11.669	Durbin-Watson:	1.218			
Prob(Omnibus):	0.003	Jarque-Bera (JB):	8.738			
Skew:	-1.326	Prob(JB):	0.0127			
Kurtosis:	5.304	Cond. No.	3.98			
=====						

Turnover from 2006–2022 was analyzed using a phase-based ordinary least squares regression, chosen because limited annual data makes complex models unreliable and phases allow clear comparison. The model explains 48% of the variation in turnover ($R^2 = 0.481$). Turnover increases significantly during the expansion phase (Phase 3), while the decline phase (Phase 2) has a negative but statistically insignificant effect. Overall, the model is statistically significant, though some residual non-normality suggests the results should be interpreted with caution.

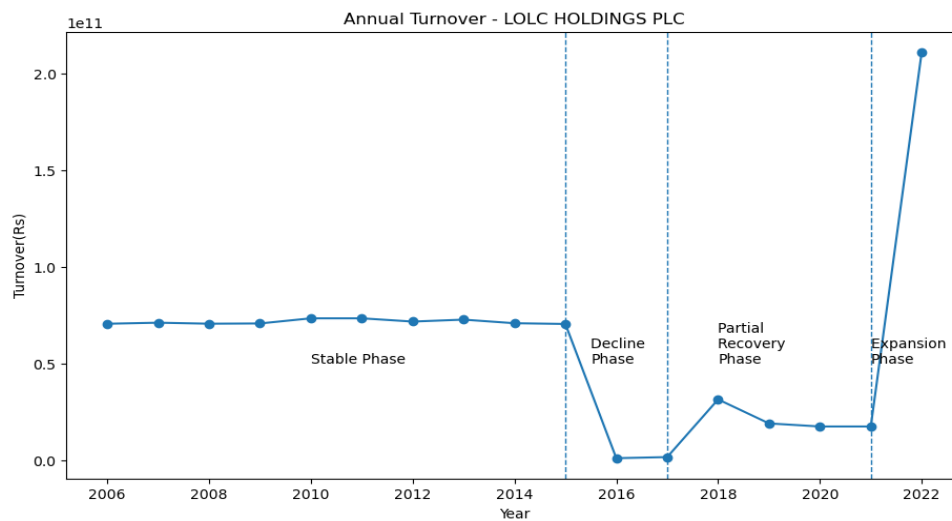
LOLC HOLDINGS PLC

Descriptive analysis

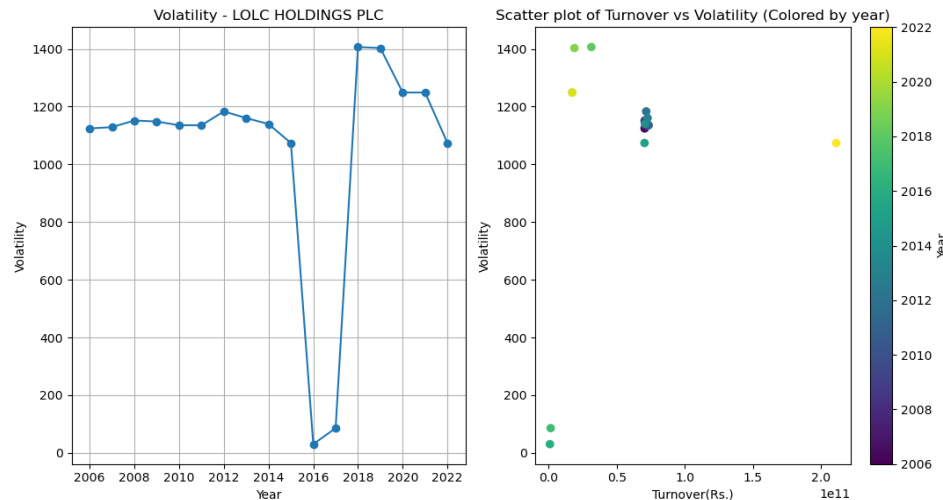
	Open_(Rs.)	High_(Rs.)	Low_(Rs.)	Close_Price_(Rs)	Trade_Volume	Share_Volume	Turnover(Rs)	Volatility
count	17.000000	17.000000	17.000000	17.000000	17.000000	1.700000e+01	1.700000e+01	17.000000
mean	299.482353	1147.382353	95.570588	557.414706	201837.941176	1.494856e+08	5.952463e+10	1051.811765
std	372.427988	404.913013	60.669481	298.764865	147245.223809	1.141729e+08	4.820011e+10	386.131896
min	72.500000	94.500000	26.500000	72.500000	3163.000000	1.079120e+07	8.951235e+08	30.400000
25%	118.000000	1210.000000	61.500000	398.000000	106838.000000	3.766430e+07	1.887274e+10	1124.500000
50%	118.000000	1210.000000	74.500000	617.300000	226526.000000	1.662745e+08	7.044131e+10	1139.800000
75%	136.000000	1210.000000	86.000000	632.500000	233927.000000	1.886108e+08	7.154487e+10	1183.500000
max	1174.500000	1489.000000	240.000000	1162.000000	675369.000000	4.923610e+08	2.109296e+11	1406.600000

Descriptive statistics for LOLC Holdings PLC indicate substantial variability across price and trading activity measures. Trading volume,

share volume, and turnover exhibit pronounced right-skewness, with median values exceeding means and maximum values far above typical levels, suggesting that a small number of years account for disproportionately high trading activity. In contrast, volatility displays relatively stable central values with limited dispersion, indicating that price uncertainty increases only during episodic events rather than persistently. Overall, the results suggest that trading behavior is characterized by generally stable conditions punctuated by short-lived periods of abnormal activity.



This figure illustrates the annual turnover of LOLC Holdings PLC over the period 2006–2022. The series exhibits a long stable phase from 2006 to 2014, during which turnover remains relatively constant with minor fluctuations. This is followed by a sharp decline in 2015–2016, indicating a significant contraction in trading activity. Although turnover partially recovers during the period 2017–2021, it remains below pre-2015 levels, suggesting an incomplete recovery. In contrast, 2022 records a substantial surge in turnover, representing an abnormal expansion rather than a continuation of a long-term trend. Overall, the figure indicates that turnover changes in distinct phases, with sudden jumps and drops rather than smooth, continuous growth.



The volatility time series shows a prolonged period of relatively stable price movements up to 2014, followed by a sharp decline in volatility during 2016–2017, indicating minimal price variation and low market activity. Volatility then rises again from 2018 onwards, reflecting

renewed uncertainty in the market. When this time pattern is examined together with the scatter plot of turnover against volatility, a clear relationship emerges. Years with extremely low volatility are consistently associated with very low turnover, suggesting weak investor participation during periods of price stagnation. However, the scatter plot also shows that exceptionally high turnover can occur even when volatility remains within its historical range, as observed in 2022. This indicates that high trading activity is not necessarily driven by large price swings but can arise from increased trading intensity and market participation, particularly during periods of heightened economic or financial stress.

Advanced Analysis

Trend Hypothesis

H_0 : There is no statistically significant trend in annual turnover over the period 2006–2022.

H_1 : There is a statistically significant trend in annual turnover over the period 2006–2022

Relationship Hypothesis

H_0 : There is no statistically significant relationship between annual turnover and key trading indicators.

H_1 : There exists a statistically significant relationship between annual turnover and key trading indicators.

Trend test (raw): Turnover(Rs)
Slope (per year): $-8.71591e+08$
p-value: 0.7274
 R^2 : 0.008338

Trend test (raw): Trade_Volume
Slope (per year): -790.826
p-value: 0.9177
 R^2 : 0.0007356

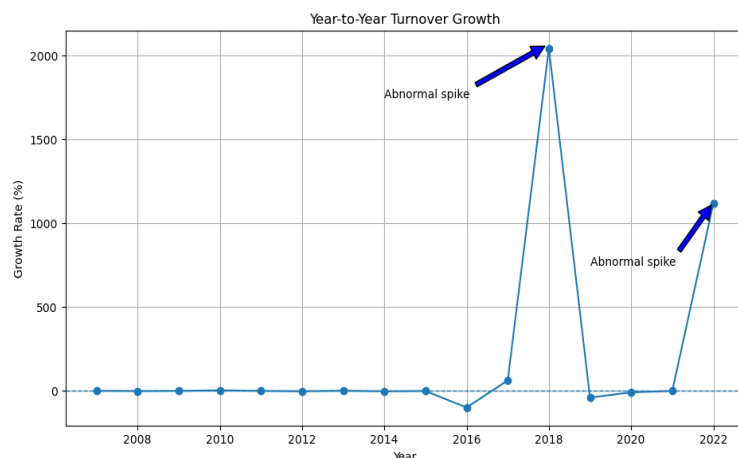
Trend test (raw): Volatility
Slope (per year): -5.30956
p-value: 0.7912
 R^2 : 0.004822

Relationship: Turnover vs Trade Volume
Spearman rho = 0.9951, p-value = $1.754e-16$
Pearson r = 0.9935, p-value = $1.462e-15$

Relationship: Turnover vs Volatility
Spearman rho = -0.0615, p-value = 0.8147
Pearson r = 0.2979, p-value = 0.2455

Linear trend analysis reveals no statistically significant long-term trends in turnover, trade volume, or volatility over the period 2006–2022, indicating that time alone does not explain variations in trading behavior. In contrast, correlation analysis using both Pearson and Spearman correlation tests shows an extremely strong and statistically significant positive relationship between turnover and trade volume, confirming that turnover is primarily driven by trading activity. No significant relationship is observed

between turnover and volatility, suggesting that price uncertainty affects trading only during exceptional periods rather than consistently. Overall, trading activity changes mainly during sudden events rather than showing steady growth over time.



The year-to-year growth rate of turnover remains close to zero for most of the study period, indicating relatively stable trading activity across consecutive years. However, two years exhibit exceptionally large positive growth rates, reflecting abnormal spikes rather than sustained growth. These spikes are preceded or followed by sharp corrections, suggesting that changes in turnover are driven by short-term shocks and low base effects rather than long-term expansion. Overall, the growth rate analysis confirms that turnover dynamics are irregular and event-driven.

Period Comparison Analysis using the Mann-Whitney U test

H_0 : The distribution (median) of the trading indicator (Turnover, Trade volume, Share volume, Volatility) is the same in the two periods 2006-2014 and 2015-2022

H_1 : The distribution (median) of the trading indicator (Turnover, Trade volume, Share volume, Volatility) differs between the two periods 2006-2014 and 2015-2022

The Mann–Whitney U test shows that turnover, trade volume, and share volume are significantly different between the periods 2006–2014 and 2015–2022, with higher trading activity in the earlier period ($p < 0.05$). This indicates that investor participation declined after 2014. However, the test does not show a significant difference in volatility between the two periods ($p > 0.05$), meaning that the range of price movements remained broadly similar. Overall, these results suggest that changes after 2014 were mainly due to lower trading activity rather than long-term changes in price behavior.

	Year	Turnover_Growth_%	Growth_Category
0	2006	NaN	NA
1	2007	0.764125	Normal fluctuation
2	2008	-0.724183	Normal fluctuation
3	2009	0.178819	Normal fluctuation
4	2010	3.762961	Normal fluctuation
5	2011	0.000000	Normal fluctuation
6	2012	-2.291395	Normal fluctuation
7	2013	1.398232	Normal fluctuation
8	2014	-2.559370	Normal fluctuation
9	2015	-0.535692	Normal fluctuation
10	2016	-98.726888	Moderate change
11	2017	63.012317	Moderate change
12	2018	2043.548822	Extreme spike
13	2019	-39.660984	Moderate change
14	2020	-8.413938	Normal fluctuation
15	2021	0.000000	Normal fluctuation
16	2022	1120.318317	Extreme spike

Variable: Turnover(Rs)
Mean (2006–2014): 7.151e+10
Mean (2015–2022): 4.604e+10
Mann-Whitney U: 63.00
p-value: 0.0107

Variable: Trade_Volume
Mean (2006–2014): 2.325e+05
Mean (2015–2022): 1.673e+05
Mann-Whitney U: 63.00
p-value: 0.0107

Variable: Share_Volume
Mean (2006–2014): 1.775e+08
Mean (2015–2022): 1.180e+08
Mann-Whitney U: 58.00
p-value: 0.0383

Variable: Volatility
Mean (2006–2014): 1.145e+03
Mean (2015–2022): 9.465e+02
Mann-Whitney U: 36.00
p-value: 1.0000

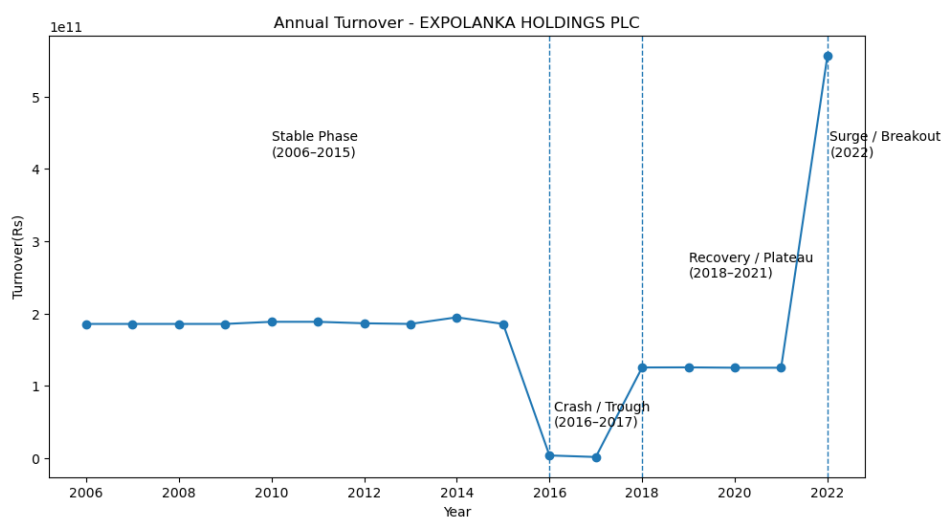
EXPOLANKA HOLDINGS PLC

Descriptive analysis

	Open_(Rs.)	High_(Rs.)	Low_(Rs.)	Close_Price_(Rs)	Trade_Volume	Share_Volume	Turnover(Rs)	Year
count	17.000000	17.000000	17.000000	17.000000	1.700000e+01	1.700000e+01	1.700000e+01	17.000000
mean	83.385294	348.547059	25.458824	232.761765	6.146304e+05	1.671127e+09	1.726120e+11	2014.000000
std	123.995957	128.469783	26.769807	135.967471	4.142041e+05	1.227824e+09	1.163108e+11	5.049752
min	6.300000	7.100000	3.800000	5.300000	4.031000e+03	2.597137e+08	1.597226e+09	2006.000000
25%	14.650000	390.000000	5.900000	182.250000	4.502060e+05	6.491184e+08	1.253677e+11	2010.000000
50%	29.300000	390.000000	29.200000	192.125000	6.593610e+05	1.829109e+09	1.855597e+11	2014.000000
75%	29.300000	390.000000	29.200000	375.250000	6.724810e+05	1.955819e+09	1.864469e+11	2018.000000
max	377.000000	405.000000	90.000000	375.250000	1.978083e+06	5.487327e+09	5.566790e+11	2022.000000

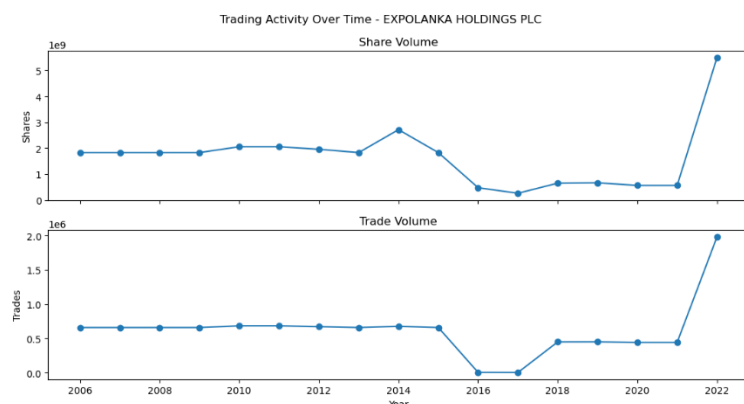
The descriptive statistics highlight the key features of EXPOLANKA HOLDINGS PLC's stock performance from 2006 to 2022. The mean opening and closing prices indicate an overall increase in share value over time, while the relatively high standard deviations reflect noticeable price volatility. The wide gap between minimum and maximum prices further confirms fluctuations in market prices. High average trade volume, share volume, and turnover demonstrate strong investor interest and active trading in the company's shares. Overall, these statistics show that EXPOLANKA HOLDINGS PLC experienced significant market activity, price variability, and growth potential during the observed period.

The annual turnover trend of EXPOLANKA HOLDINGS PLC shows distinct phases over the period from 2006 to 2022. From 2006 to 2015, turnover remained relatively stable, indicating consistent trading activity. A



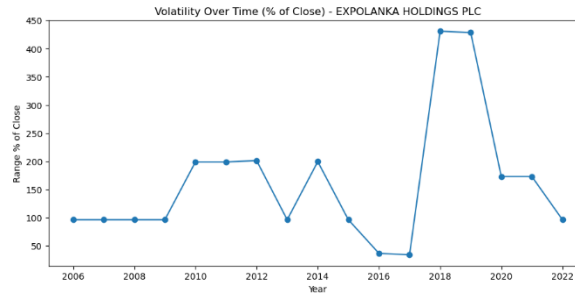
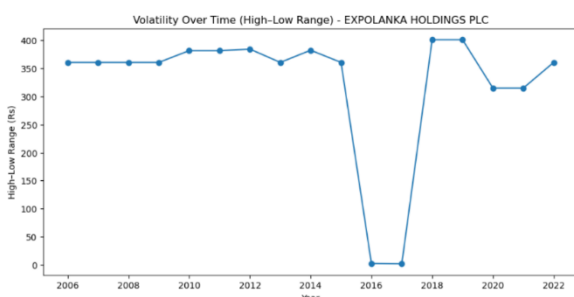
sharp decline is observed during 2016–2017, representing a significant market downturn. This was followed by a recovery and plateau phase from 2018 to 2021, where turnover stabilized at a moderate level. In 2022, the company experienced a dramatic surge in turnover, suggesting a strong

market breakout and renewed investor interest. Overall, the pattern reflects periods of stability, disruption, recovery, and rapid growth in trading activity.

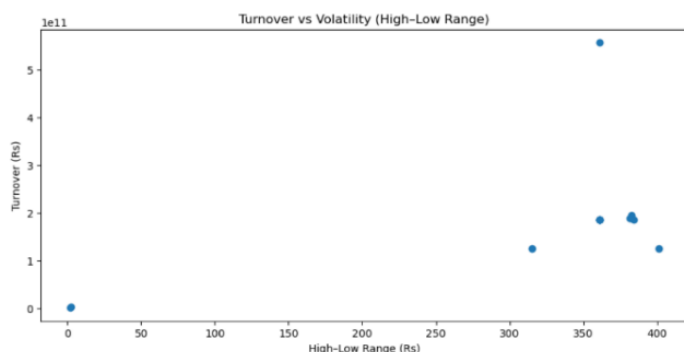


The trading activity of EXPOLANKA HOLDINGS PLC, measured through share volume and trade volume, shows distinct patterns over the period from 2006 to 2022. From 2006 to around 2014, both share volume and trade volume remained relatively stable, indicating consistent market participation. A noticeable decline occurs during 2016–2017, where both volumes drop sharply, reflecting a

significant slowdown in trading activity. From 2018 onwards, a gradual recovery is observed, with volumes stabilizing at moderate levels. In 2022, both share volume and trade volume rose sharply, highlighting a surge in market activity and heightened investor interest. Overall, the trends reflect periods of stability, contraction, recovery, and strong expansion in EXPOLANKA’s trading behavior.

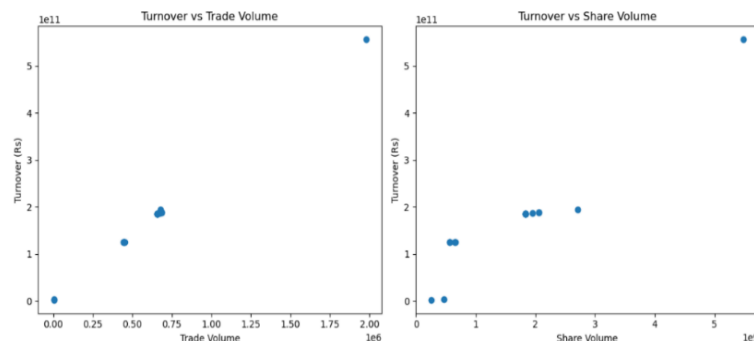


The volatility analysis of EXPOLANKA HOLDINGS PLC illustrates fluctuations in price movements over time using the high–low price range and the range as a percentage of the closing price. From 2006 to around 2015, volatility remained relatively high but stable, indicating regular price fluctuations within a consistent range. A sharp decline in volatility is observed during 2016–2017, reflecting reduced price movement and lower market activity during this period. In contrast, volatility spikes significantly in 2018–2019, highlighting increased uncertainty and heightened market reactions. From 2020 onwards, volatility moderates but remains noticeable, suggesting continued price fluctuations with comparatively more stability. Overall, the analysis reveals changing risk levels over time, with periods of stability, low activity, sharp uncertainty, and subsequent normalization in EXPOLANKA’s stock behavior.



The scatter plots illustrate the relationship between turnover and key trading indicators of EXPOLANKA HOLDINGS PLC. Turnover shows a strong positive association with both trade volume and share volume, indicating that higher trading activity leads to increased turnover. In contrast, the relationship between turnover and price volatility (measured by the high–low range) is weaker and more

dispersed, suggesting that large price fluctuations do not consistently translate into higher turnover. Overall, these plots confirm that turnover is primarily driven by trading activity rather than by price volatility.



Advanced Analysis Hypothesis Testing

- **Null Hypothesis (H_0)** -: There is no statistically significant trend or relationship between Year and key annual trading indicators (Turnover, Trade Volume, Share Volume).
- **Alternative Hypothesis (H_1)** -There exists a statistically significant trend or relationship between Year and key annual trading indicators.

Since $p > 0.05$ for all variables $\rightarrow$ Fail to reject H_0

Interpretation: The Pearson correlation analysis shows no statistically significant linear relationship between Year and the key annual trading indicators of EXPOLANKA HOLDINGS PLC. The correlation coefficients are very close to zero, indicating extremely weak associations, and the high p-values confirm that these relationships are not statistically significant. Therefore, the null hypothesis is not rejected, suggesting that changes in turnover, trade volume, and share volume do not follow a consistent linear trend over time.

Turnover(Rs): Pearson $r = 0.075$, p-value = 0.7755
 Trade_Volume: Pearson $r = 0.071$, p-value = 0.7863
 Share_Volume: Pearson $r = -0.071$, p-value = 0.7876

Trend Significance using Linear Regression

The linear regression analysis was used to assess whether a significant time trend exists between year and the key trading indicators of EXPOLANKA HOLDINGS PLC, namely turnover, trade volume, and share volume. Although turnover and trade volume exhibit small positive slope coefficients and share volume shows a slight negative slope, none of these trends are statistically significant, as all p-values are well above the 5% significance level. The extremely low R-squared values across all models indicate that year explains very little of the variation in these trading indicators. Consequently, the results suggest that there is no statistically significant linear trend in turnover, trade volume, or share volume over time, and the null hypothesis of no significant trend is not rejected.

Trend Model: Turnover(Rs) ~ Year
 Slope (Year coefficient): 1722438926.46737
 p-value: 0.7754561678074653
 R-squared: 0.0055922629389485445

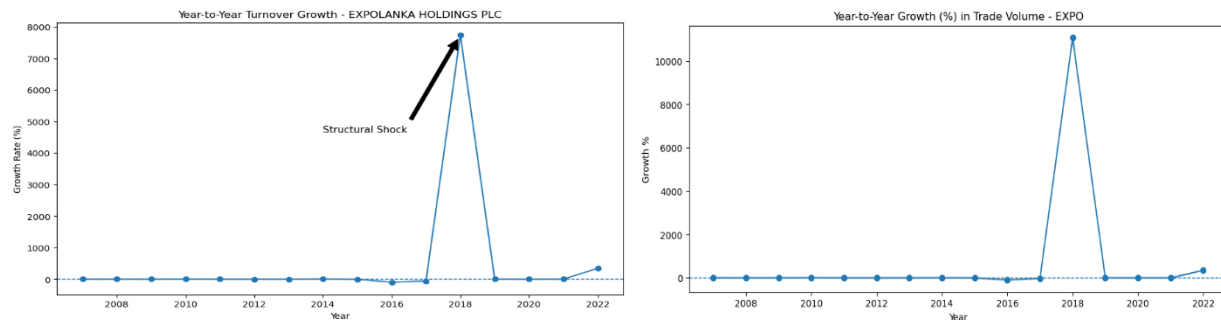
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Trend Model: Trade_Volume ~ Year
 Slope (Year coefficient): 5830.965686275238
 p-value: 0.7862985582322002
 R-squared: 0.005053502058579129

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Trend Model: Share_Volume ~ Year
 Slope (Year coefficient): -17174135.96568691
 p-value: 0.7876362978876761
 R-squared: 0.004989045996220298

Growth rate analysis



The year-to-year turnover growth remains relatively stable across most years, reflecting consistent performance. However, a sharp spike around 2018 indicates a significant structural shock and an exceptional increase in turnover. After this surge, growth rates return to more moderate levels, suggesting a normalization of performance. Overall, the analysis shows stable growth with one notable period of extraordinary expansion.

The year-to-year growth rate of trade volume remains relatively stable and close to zero across most years, indicating consistent trading activity. A sharp and extreme spike is observed around 2018, representing an exceptional surge in the number of trades and a structural shock to normal trading patterns. Following this peak, growth rates return to moderate levels, suggesting a normalization of trade activity. Overall, the analysis highlights generally stable trade volume growth with one significant period of extraordinary expansion.

Period Comparison Analysis (Mann–Whitney U test (Two-sample non-parametric test))

Null Hypothesis (H_0): There is no statistically significant difference in the median annual trading indicator (e.g., turnover) between the two sub-periods.

Alternative Hypothesis (H_1): There is a statistically significant difference in the median annual trading indicator between the two sub-periods.

The Mann–Whitney U test was used to compare the median annual turnover between two sub-periods: **2006–2015** and **2016–2022**. This non-parametric test is appropriate as it does not assume normality and assesses whether the central tendency of turnover differs between the two periods. The test results show a Mann–Whitney U statistic of **60.0** with a **p-value of 0.0144**, which is below the 5% significance level. Therefore, the null hypothesis is rejected, indicating a **statistically significant difference in median annual turnover** between the two sub-periods. This suggests that EXPOLANKA's turnover performance changed meaningfully after 2015, reflecting a structural shift in trading activity in the later period.

Mann-Whitney U statistic: 60.0
p-value: 0.014401972890686614

Conclusion

The analysis of annual trading statistics for the four selected companies reveals that turnover behavior is irregular and non-linear across the study period, with clear evidence of distinct phases, sharp fluctuations, and occasional extreme values. These patterns suggest that turnover does not evolve smoothly over time and cannot be adequately explained by a simple long-term trend. Such irregular behaviors are likely influenced by changes in overall market conditions, shifts in investor participation, and episodic surges in trading activity, which can cause sudden expansions or contractions in turnover even when long-run trends remain unclear.

A common pattern across all four companies is that turnover changes are closely associated with changes in trading activity, particularly share volume and trade volume. Correlation and regression results consistently indicate that increases in turnover tend to occur alongside increases in trading intensity, highlighting liquidity-related behavior as a dominant feature of market activity. In contrast, volatility does not show a uniform or stable relationship with turnover across all firms, indicating that price dispersion alone is not a consistent driver of trading value.

Despite these similarities, company-level differences are evident. Some firms exhibit highly concentrated growth periods and extreme turnover spikes, while others show relatively smoother and more stable patterns. These differences suggest that, although companies are exposed to common market conditions, the magnitude and structure of turnover responses vary across firms, likely reflecting differences in trading behavior and market participation.

Phase-based analysis further indicates that visual differences between periods do not always translate into statistically significant differences, particularly when using non-parametric tests. This reflects the combined effect of limited annual observations and high variability within phases. Overall, the findings emphasize that turnover in the selected companies is driven more by shifts in trading activity and structural changes than by gradual trends, reinforcing the importance of phase-aware and company-specific analysis when examining long-term trading behavior.

Limitations

This study is subject to several limitations. The use of annual data limits the ability to capture short-term fluctuations and market reactions within each year. The analysis is based on a small sample of companies, which restricts the generalization of findings to the broader Colombo Stock Exchange. Market phases were identified through observed patterns rather than formal structural break tests, introducing potential subjectivity. Finally, the analysis focuses on trading statistics alone, without incorporating external economic or firm-level explanatory variables, which limits deeper interpretation of the underlying causes of observed turnover changes.